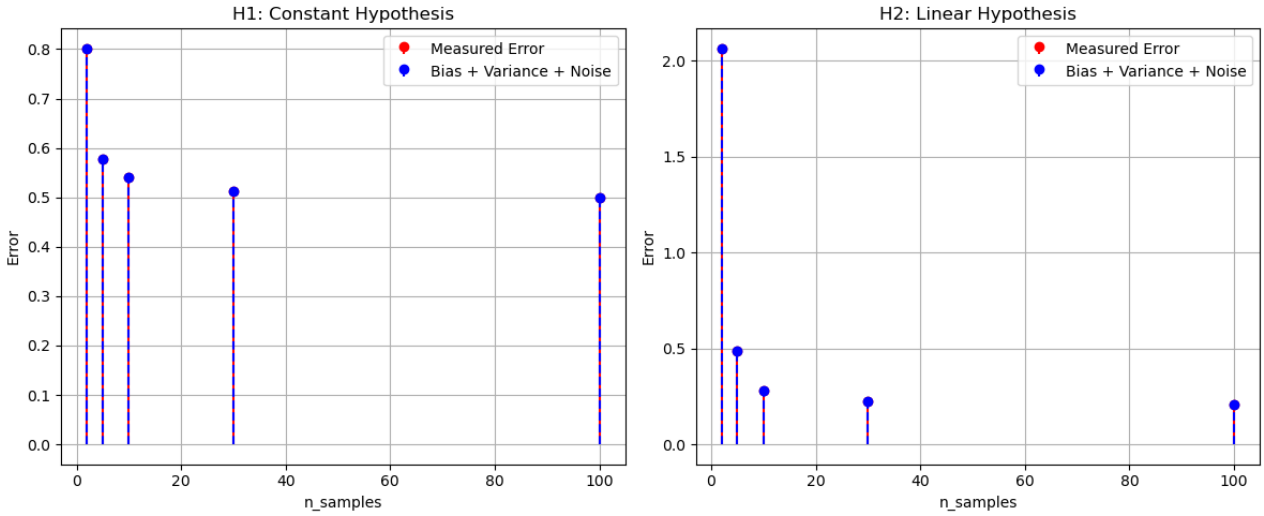

a)
According to the overall error from the given curves it is seen that for n=2, H1hypothesis works better than H2.

b) When n change from n=2 to n=10,
i) for H1, Not much change in all the parameters was observed. The measured error converges at approx. 0.5.

ii) for H2, the error dropped drastically to approx. 0.25.

```
cse2203320@ltsp129: ~/Desktop/lab
.2387, Var(H2)=1.8911, Error(H2)=2.1298, Noise=0.0000
n_samples=5: Bias(H1)=0.4958, Var(H1)=0.1044, Error(H1)=0.6002, Bias(H2)=0
.2083, Var(H2)=0.1870, Error(H2)=0.3954, Noise=0.0000
n_samples=10: Bias(H1)=0.4951, Var(H1)=0.0570, Error(H1)=0.5520, Bias(H2)=
0.2055, Var(H2)=0.0748, Error(H2)=0.2803, Noise=0.0000
n_samples=30: Bias(H1)=0.4951, Var(H1)=0.0159, Error(H1)=0.5111, Bias(H2)=
0.2041, Var(H2)=0.0140, Error(H2)=0.2181, Noise=0.0000
n_samples=100: Bias(H1)=0.4951, Var(H1)=0.0055, Error(H1)=0.5005, Bias(H2)
=0.2041, Var(H2)=0.0044, Error(H2)=0.2085, Noise=0.0000
n_samples= 2: Bias(H1)=0.4951, Var(H1)=0.2706, Error(H1)=0.7657, Bias(H2)=
0.5568, Var(H2)=14.5046, Error(H2)=15.0614, Noise=0.0448
n_samples= 2: Bias(H1)=0.5048, Var(H1)=0.1984, Error(H1)=0.7032, Bias(H2)=
0.2314, Var(H2)=1.7562, Error(H2)=1.9876, Noise=0.0000
cse2203320@ltsp129:~/Desktop/lab$
```

c) For noiseless, H2 had better bias then H1 whereas variance of H1 was better than H2.
For noisy case,H1’s bias and variance was better than H2’s.